# World Models for Autonomous Driving: From Future Generation to Decision Making

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***Abstract*—Autonomous driving systems must anticipate traffic evolution, compare candidate actions, and operate safely in closed loop. Recent world models address this need by generating future scenes, rolling out structured states, or exposing transitions for decision making. This survey takes a model predictive control (MPC) perspective. We treat driving world models as transition interfaces for action-conditioned rollout, cost exposure, constraint handling, uncertainty representation, and receding-horizon replanning. The literature is organized into future world generation, planning with world models, and hybrid prediction–planning systems. Across these roles, we compare visual, bird's-eye-view, vectorized, occupancy, LiDAR, and latent states with their planner-facing interfaces. We also synthesize datasets, simulators, metrics, benchmark protocols, computational constraints, and deployment evidence. A protocol-controlled table reports comparable results without ranking incompatible settings. The review identifies open problems in long-horizon reliability, multimodal representation, generation-to-planning transfer, efficient deployment, and foundation-model-assisted safety validation. Predicted futures should ultimately be judged by the decisions they support, not by realism alone. Resources are available at the project page.**

***Index Terms*—Autonomous driving, world models, model predictive control, planning, simulation, closed-loop evaluation.**

## I. Introduction

AUTONOMOUS driving requires more than accurate recognition of the current traffic scene. A driving system must also anticipate how the environment may change under different actions, interactions, and constraints. We use *driving world model* for a learned representation that can be rolled forward in time. Its future states may support generation, simulation, evaluation, or decision making. Unlike perception or conventional agent forecasting, it models scene evolution and may expose action-conditioned consequences to a planner.

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World models now span sensor synthesis, structured scene rollout, planning, and interactive simulation [1]–[3]. Definitions and evaluation protocols remain inconsistent. A focused review must distinguish what each model predicts, how a planner consumes it, and which claims rest on open-loop, non-reactive, or reactive evidence. This motivation rests on the technical scope and evidence gap documented in the cited literature, not an unreproducible publication-count trend.

Methods differ in representation and mechanism, but their role in the driving pipeline is more consequential. Some generate futures, others evaluate trajectories, and a smaller group shares prediction with planning or closed-loop simulation. We therefore organize the field by task role rather than appearance or architecture alone.

### A. Comparison to Existing Surveys

Existing surveys cover world models, video and 3-D generation, connected autonomous vehicles, occupancy, foundation models, and scenario generation [1]–[12]. Motion-planning and MPC surveys instead examine constraints, receding-horizon optimization, and vehicle control [13], [14]. A recent survey also examines latent world models for autonomous driving and their integration with MPC [15]. We examine planner-facing world-model interfaces through action-conditioned rollout, costs, constraints, uncertainty, and replanning. Table I summarizes the organizing axes of adjacent survey lines.

The same predicted future can help or mislead, depending on the interface. Video favors inspection, BEV and occupancy expose geometry, and latents allow cheap rollout. Planning evidence must also account for feedback after the ego vehicle acts. Recent benchmarks therefore test actionability and repeated execution, with different degrees of agent reactivity [16]–[20].

### B. Distinction, Contributions, and Roadmap

Our taxonomy separates generating a future from using it. It also distinguishes offline prediction from interactive simulation, where executed actions change the next world state.

The contributions of this survey are summarized as follows:

- We define a primary-role taxonomy—future world generation, planning with world models, and hybrid prediction–planning—with explicit boundary and secondary-capability rules.
- We operationalize the MPC perspective through five evidence elements: action-conditioned rollout, cost exposure,

TABLE I
POSITIONING RELATIVE TO ADJACENT SURVEY LINES. ENTRIES DESCRIBE EACH SURVEY'S PRIMARY EMPHASIS; A BLANK CIRCLE INDICATES THAT THE DIMENSION IS NOT A CENTRAL ORGANIZING AXIS, NOT THAT THE TOPIC IS ABSENT.

| Survey line | Primary emphasis | AD-specific | Task-role taxonomy | MPC interface | Evidence ladder | Quantitative protocol control |
|---|---|---|---|---|---|---|
| General world-model surveys [4] | Concepts, agents, and cross-domain applications | ○ | ○ | ○ | ○ | ○ |
| Embodied-AI WM survey [6] | Function, time, space, and embodied-agent uses | ○ | ○ | ○ | ○ | ○ |
| Initial AD-WM survey [1] | Early concepts and application categories | ● | ○ | ○ | ○ | ○ |
| AD-WM survey [2] | Generation, planning, interaction, and components | ● | ● | ○ | ○ | ○ |
| DWM ecosystem survey [3] | Modalities, applications, datasets, and ecosystem roles | ● | ○ | ○ | ○ | ○ |
| Generation surveys [7]–[9] | Video and physical-future generation | ○ | ○ | ○ | ○ | ○ |
| CAV WM survey [10] | Cooperative and infrastructure-assisted WMs | ● | ○ | ○ | ○ | ○ |
| Occupancy perception survey [11] | Occupancy representation, fusion, and supervision | ● | ○ | ○ | ○ | ○ |
| FM scenario survey [12] | Foundation-model scenario generation and testing | ● | ○ | ○ | ● | ○ |
| Planning and MPC surveys [13], [14] | Dynamics, constraints, optimization, and control | ● | ○ | ● | ○ | ○ |
| **Our survey** | **AD-WM roles, interfaces, evidence, and deployment** | ● | ● | ● | ● | ● |

●: central organizing axis; ○: not a central organizing axis. The comparison concerns analytical organization, not overall survey quality or coverage.

constraint handling, uncertainty representation, and replanning. Tables II–IV apply this rubric to representative methods.

- We connect representation and mechanism choices to computation, deployment, safety-critical fidelity, and an evaluation ladder spanning open-loop, non-reactive, reactive, and higher-fidelity evidence. Table V adds a protocol-controlled quantitative synthesis.
- We derive testable research directions for long-horizon reliability, multimodal state representation, generation-to-planning transfer, efficient deployment, and foundation-model-assisted safety validation.

Fig. 1 links the definition and taxonomy to comparison, evaluation, and open challenges. Sections II–VII follow this sequence.

## II. DRIVING WORLD MODELS: SCOPE AND MPC-ORIENTED INTERFACE

### A. Definition

We define a world model as a learned, temporally evolving representation of the driving environment. It can be conditioned, rolled forward, and consumed by simulation or decision modules. The state may be visual, BEV, occupancy, point-cloud, vectorized, or latent. Conditions may include observations, ego actions, trajectories, maps, language, or scene metadata. Short-horizon predictors without rollout and policies without an explicit world representation fall outside our scope.

Visual plausibility alone is insufficient. Driving rollouts must preserve relevant topology, geometry, interaction, and action effects. Conversely, a useful model need not render photorealistic images; structured or latent states qualify when they support controlled rollout and decision-relevant evaluation.

We compare models through a learned transition interface that advances an explicit or latent state and exposes predicted consequences. This abstraction makes heterogeneous representations comparable without treating them as interchangeable.

### B. Formal MPC-Oriented Interface

Let $\mathbf{s}_t$ be the world state, $\mathbf{a}_t$ an ego action or motion primitive, and $\mathbf{z}_t$ exogenous context. A generic stochastic transition is

$$\begin{aligned} \mathbf{s}_{t+1} &\sim \hat{p}_\theta(\cdot \mid \mathbf{s}_t, \mathbf{a}_t, \mathbf{z}_t), \\ \mathbf{o}_{t+1} &\sim \hat{p}_\theta(\cdot \mid \mathbf{s}_{t+1}, \mathbf{z}_{t+1}), \end{aligned} \tag{1}$$

where $\mathbf{o}_{t+1}$ is an optional observation-level output. Deterministic, diffusion, autoregressive, ensemble, and latent-dynamics models instantiate different forms of $\hat{p}_\theta$.

For horizon $H$, an MPC-style planner evaluates candidate action sequences through

$$\begin{aligned} \mathbf{a}^\star_{0:H-1} = \arg\min_{\mathbf{a}_{0:H-1}} \mathbb{E}_{\hat{p}_\theta}\left[\sum_{k=0}^{H-1} \ell(\mathbf{s}_k, \mathbf{a}_k) + \ell_T(\mathbf{s}_H)\right], \\ \text{s.t. } \mathbf{s}_{k+1} \sim \hat{p}_\theta(\cdot \mid \mathbf{s}_k, \mathbf{a}_k, \mathbf{z}_k), \\ h(\mathbf{s}_k, \mathbf{a}_k) \le 0, \quad k = 0, \dots, H-1, \end{aligned} \tag{2}$$

where $\ell$ and $\ell_T$ are running and terminal costs, and $h$ represents feasibility or safety constraints. Equation (2) is an analytical reference rather than a claim that every reviewed method explicitly solves MPC. It decomposes planner-facing world

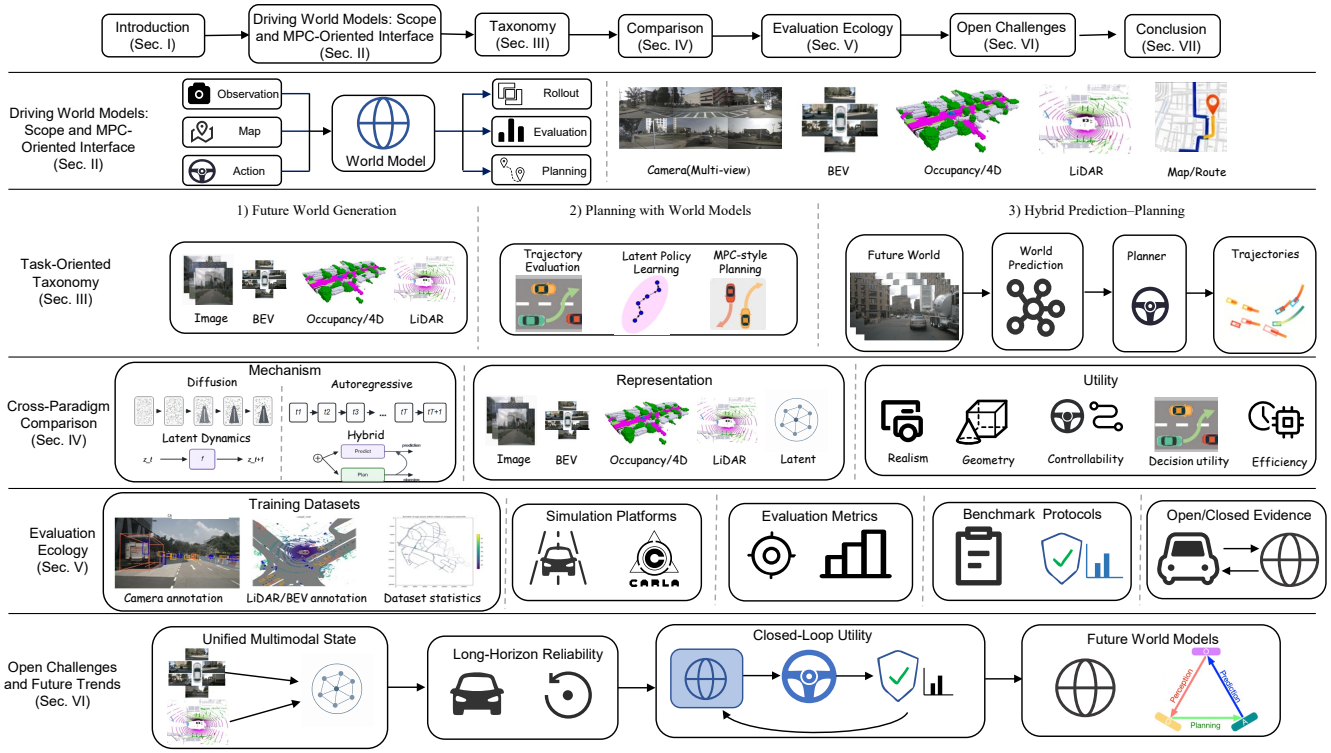



Fig. 1. Survey roadmap from the predictive transition interface to taxonomy, comparison, evaluation, and open challenges. Formal category boundaries are defined in Section III.

modeling into five interface elements that can be inspected across heterogeneous methods. We denote these elements by A/C/H/U/R: action-conditioned rollout (A), cost exposure (C), constraint handling (H), uncertainty representation (U), and replanning evidence (R).

*Action-conditioned rollout* asks whether a future changes under candidate ego actions. *Cost exposure* records rewards, values, risks, progress, comfort, or rule compliance that can define $\ell$ or $\ell_T$. *Constraint handling* checks whether geometry, collisions, traffic rules, or dynamics enter generation, filtering, or optimization. *Uncertainty representation* must be decision-usable, for example by changing a safety margin or fallback rule. Finally, *replanning evidence* requires prediction and decision to repeat after execution produces a new observation. Log replay and one-shot trajectory scoring remain open loop even when they evaluate a multi-step horizon.

Codes are explicit (E), partial (P), or not demonstrated (–), with per-entry evidence in Supplementary Tables S1–S3. E requires an implemented interface or reported evaluation; P is indirect support; – means insufficient evidence, not impossibility. C=E requires an exposed cost/reward or selection score, rather than a generic loss or benchmark score. U=E requires uncertainty to change a decision rule. H=E includes explicit penalties/filtering, with agent and phase specified.

A representative architecture follows the same flow. Current observations, map, and route context form $\mathbf{s}_t$. Candidate ego actions query $\hat{p}_\theta$ to obtain multi-step state distributions. A decision module converts those rollouts into costs and constraint checks, selects an action sequence, executes its first action, and repeats after the next observation. Future-generation models provide states; planning-oriented models add a separable decision interface. Hybrid models share prediction and action, while interactive simulators also update the environment. This architecture anchors Section III.

Fig. 2 visualizes this representative information flow.

### C. Relation to Prediction, Simulation, and Planning

World models overlap with prediction, simulation, and planning. They extend agent forecasting toward scene-level state transitions, but they do not by themselves define the simulator or final control objective.

Motion forecasters provide multi-agent futures, map context, and uncertainty that can support world models [21]–[26]. Traffic-risk studies add decision variables that a world model may expose. Connected-vehicle trajectory prediction supports cut-in warning, potential fields quantify lane-change risk, and multi-task learning jointly predicts lane-change probability, time, and collision-risk level [27]–[29]. These components qualify only when embedded in a scene transition under intervention. We apply the same rule to end-to-end planners [30]–[32].

### D. Scope, Category Boundaries, and Comparison Dimensions

The survey assigns each method by its *primary task role*. *Future world generation* produces future observations or states without requiring planning in the main objective. *Planning with world models* consumes predicted states, transitions,

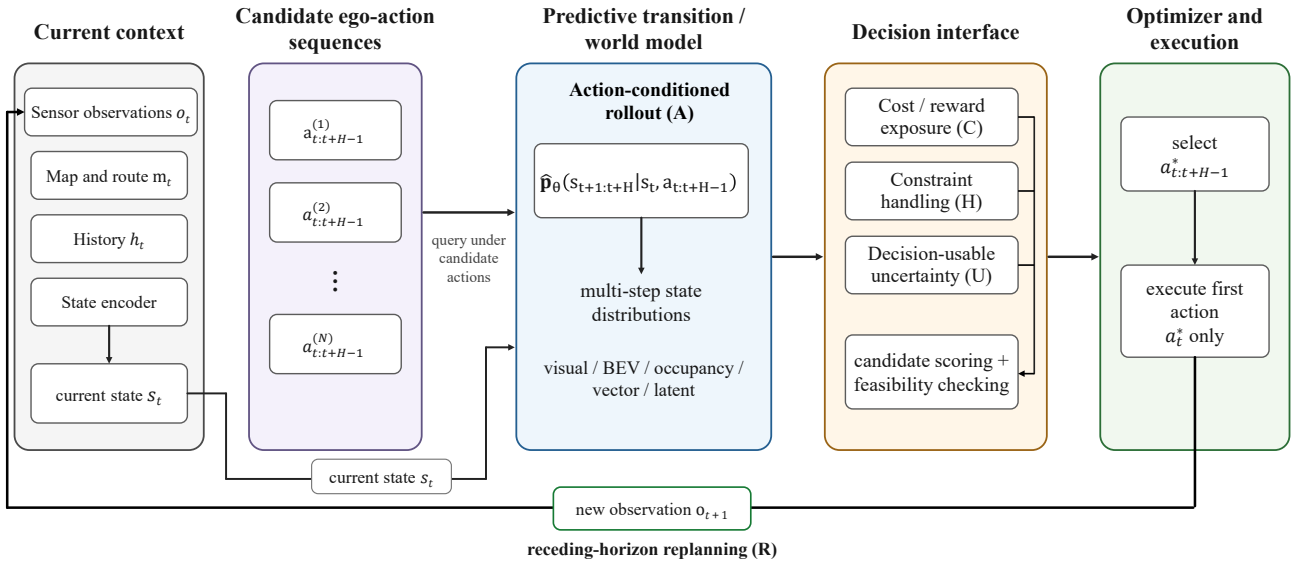



Fig. 2. Representative MPC-oriented information flow for a driving world model. Candidate ego-action sequences query a predictive transition; multi-step state distributions expose costs or rewards, constraints, and decision-usable uncertainty; the optimizer executes only the first selected action before replanning from the next observation. The diagram is a reference architecture; individual methods support different subsets of the five MPC interface elements.

rewards, or values through a separable scorer, optimizer, or policy. *Hybrid prediction–planning* shares state, objective, or executable interaction between prediction and decision. A visual output alone does not make a method hybrid, and a BEV or latent output alone does not make it planning oriented.

Task role is a continuum, so a method may have secondary capabilities outside its primary category. Each method is assigned to one primary category according to its stated objective, system dependency, and strongest evaluation evidence; secondary capabilities are recorded separately. Planning-primary methods evaluate trajectory selection as a central function. Hybrid methods remove part of the separation between prediction and decision. Boundary cases are stated in the text and tables.

The assignment can be reproduced with three questions. First, can the learned transition be evaluated without a decision module? Second, does trajectory or action selection form a central reported outcome? Third, do prediction and decision share parameters, supervision, or executable feedback that cannot be removed without changing the method? A positive answer only to the first question indicates generation. The second indicates planning-primary use, whereas the third provides hybrid evidence. This procedure prevents output modality from deciding the category.

Pure perception, forecasting without world-state rollout, policies without a world representation, and language-only reasoning remain enabling components rather than primary world models.

We compare task role, representation, conditions, mechanism, output interface, and evaluation protocol. Together, these dimensions show what the state contains, how it evolves, how another module consumes it, and what evidence supports the claimed role.

## III. Task-Oriented Taxonomy

Fig. 3 organizes methods by primary task role and then by representation or mechanism. The downward coupling arrow denotes increasing integration between the learned transition and decision modules. Generation exposes a future state, planning consumes it through a separable interface, and hybrid systems share representations, objectives, or execution. The arrow does not indicate greater visual fidelity or dimensionality.

Tables II–IV compare 29 temporal world models: nine generation-primary, ten planning-primary, and ten hybrid systems. RangeLDM is an additional static sensor-generation boundary comparator, giving 30 audited rows. All tables record subclass/state, transition or decision interface, A/C/H/U/R, and evaluation scope; Supplementary Tables S1–S3 identify the supporting primary-source evidence. Each temporal model has one primary row. The vector/graph cross-reference in Table II points to VectorWorld in Table IV and is not counted twice.

### A. Future World Generation

Future world generation asks how to represent and roll forward the environment. Table II separates controllable generation from decision use: GAIA-1 and DriveDreamer have explicit A, while C and R are not demonstrated in their listed evaluations [35], [36]. Their A entries support counterfactual scene generation. A direct MPC use would additionally need a cost interface and execution feedback. Representation fidelity must therefore be judged against the interface that consumes the rollout.

Fig. 4 maps trade-offs in realism, geometry, sensor fidelity, and planning compatibility. A representation becomes planning oriented only when a decision module queries its rollout.

*1) Visual World Models:* Visual models generate future sensor observations that are easy to inspect but costly to

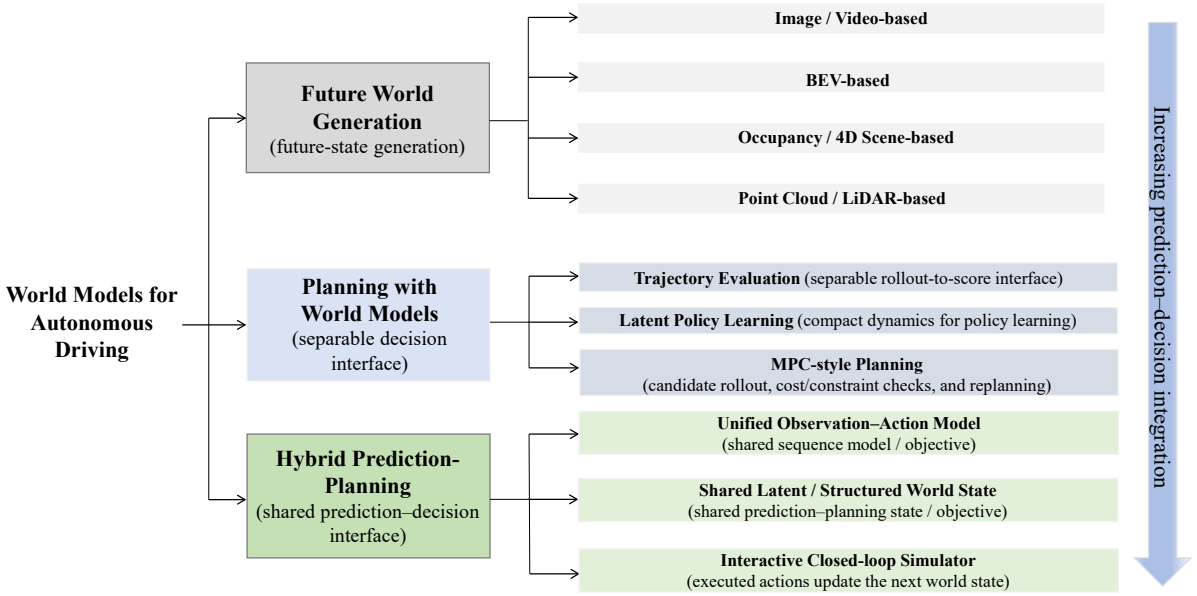



Fig. 3. A task-oriented taxonomy of autonomous-driving world models based on each method's primary task role: future world generation, planning with world models, or hybrid prediction–planning. The downward arrow denotes increasing integration between learned transition and decision modules, not increasing visual fidelity, modality richness, or representation dimensionality.

TABLE II
REPRESENTATIVE AUTONOMOUS-DRIVING METHODS (PART 1: FUTURE WORLD GENERATION AND A STATIC BOUNDARY CASE)

| Study | Subclass / State | Transition and Output | MPC Evidence A/C/H/U/R | Evaluation Evidence |
|---|---|---|---|---|
| FIERY [33] (ICCV'21) | BEV prediction | Probabilistic BEV dynamics; future instances, motion, and trajectories. | –/–/–/P/– | nuScenes/Lyft; open-loop future prediction and multimodality. |
| BEVWorld [34] (arXiv'25) | BEV / diffusion | Ego-motion-conditioned BEV tokens; joint camera and LiDAR futures. | E/–/–/P/– | nuScenes/CARLA; sensor forecasting and perception/motion-prediction transfer. |
| GAIA-1 [35] (arXiv'23) | Visual / AR | Video, text, and ego-action tokens; future camera video. | E/–/P/P/– | Internal driving data; controllable visual generation, without planning evidence. |
| DriveDreamer [36] (ECCV'24) | Visual / diffusion | HD-map, box, and ego-action conditions; future video and layout. | E/–/P/P/– | nuScenes; generation quality, controllability, and perception augmentation. |
| DriveDreamer-2 [37] (AAAI'25) | Visual / diffusion | Prompt-derived trajectories and layouts; customized multi-view video. | P/–/P/P/– | nuScenes; visual quality, rare-scene control, and perception augmentation. |
| Copilot4D [38] (ICLR'24) | LiDAR / diffusion | Ego-pose-conditioned point-cloud tokens; stochastic future point clouds. | E/–/–/P/– | nuScenes/KITTI/AV2; open-loop Chamfer-distance forecasting. |
| OccWorld [39] (ECCV'24) | Occupancy / AR | Joint future occupancy and ego-trajectory tokens. | P/–/P/P/– | nuScenes; occupancy forecasting and open-loop planning metrics. |
| DOME [40] (arXiv'24) | Occupancy / diffusion | Controllable multi-step occupancy generation. | E/–/P/P/– | nuScenes; long-horizon occupancy fidelity and controllability. |
| RangeLDM (boundary) [41] (ECCV'24) | Static LiDAR / diffusion | Range-image generation, upsampling, and inpainting. | –/–/–/–/– | KITTI-360/nuScenes; single-scan synthesis, not temporal rollout. |
| SparseWorld [42] (AAAI'26) | Occupancy / sparse | Ego-state-conditioned sparse queries; 4-D occupancy futures. | P/–/P/–/– | Perception, forecasting, planning, and efficiency; no reported receding-horizon test. |
| *VectorWorld (cross-ref.)* | Vector/graph | Lane–agent graph tiles; reactive non-ego motion. | Primary profile and evaluation: Table IV. | |

*MPC evidence:* A, action-conditioned rollout; C, cost/reward; H, constraints; U, decision-usable uncertainty; R, replanning. E: explicit; P: partial; –: not demonstrated. Supplementary Table S1 gives source anchors and scoped rationales. RangeLDM is a static sensor-generation comparator, excluded from temporal-world-model counts. The vector/graph row cross-references Table IV without double counting.

convert into planning variables. GAIA-1 predicts unified video, text, and action tokens, then decodes video through diffusion [35]. DriveDreamer conditions diffusion on maps, boxes, and text; DriveDreamer-2 first induces trajectories and layouts from prompts [36], [37]. GAIA-2 adds a continuous tokenizer, multi-camera consistency, editing, and rollout [43]. STAGE

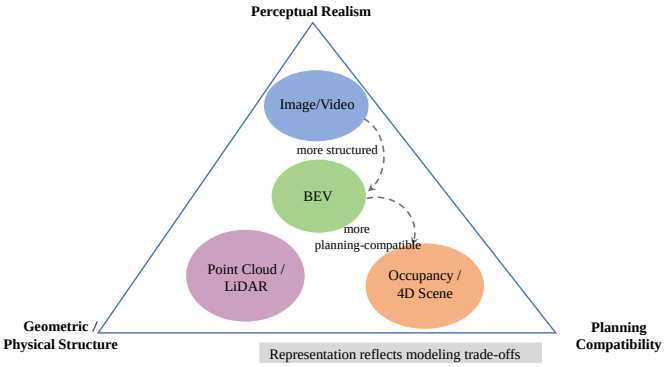



Fig. 4. Representation-space trade-offs for future world generation. Visual, BEV, occupancy, and LiDAR world models expose different balances among perceptual realism, geometric structure, sensor fidelity, and planning compatibility.

and Orbis address streaming generation and longer-horizon trajectory realism [44], [45].

These models are most useful when appearance, editing, or human inspection matters. Yet a realistic video can miss the commanded lane change or interaction. It can also violate multi-camera geometry. ACT-Bench and WorldSimBench therefore test action following and action-level usefulness in addition to visual quality [19], [46].

Planner use requires a second layer of evidence. The generated scene must expose route progress, collision geometry, and the consequence of each candidate maneuver. Multi-camera consistency should also be tested after ego motion, not only at the logged viewpoint. Otherwise, visual quality can improve while the decision interface remains unchanged.

*2) BEV World Models:* BEV aligns multi-view observations on the ground plane and preserves road topology, object layout, and motion. FIERY and PowerBEV predict future instances in this space [33], [47]. StretchBEV adds stochastic residual dynamics, while CarFormer uses object-centric slots [48], [49]. MILE couples BEV latent dynamics to policy learning, and BEVWorld rolls out action-conditioned BEV tokens [34], [50].

BEV is compact and planner-friendly, but loses vertical structure and appearance cues. It builds on BEV perception methods that lift images, depth, temporal context, or multiple sensors into a common state [51]–[54]. World modeling adds a harder requirement: the state must remain coherent after ego and agent motion.

This coherence is inspectable. Centers, velocities, lane relations, route corridors, dynamic occupancy, and cost maps can be checked at every rollout step. Height, shape, traffic-light appearance, and subtle visual cues are less visible. BEV systems should therefore state which missing variables are recovered elsewhere in the planning stack.

*3) Vectorized and Graph-Structured World States:* Vector states encode lanes, routes, controls, and agents as polylines and relations. VectorNet established this representation for forecasting, while VAD uses vectorized agents and maps as planning constraints [21], [32]. These methods provide important state and constraint interfaces, but a vectorized planner or predictor is not automatically a world model. It must also learn how the structured state evolves under time and intervention.

VectorWorld generates lane–agent graph tiles and couples them to a reactive non-ego policy [55]. Table II lists this vector/graph interface and points to its primary hybrid entry in Table IV. Its executed feedback loop and explicit non-ego kinematic shaping support R=E and H=E, respectively; these simulator constraints do not guarantee ego safety. Vector states avoid dense rendering but depend on map and perception quality and can omit unexpected objects or appearance cues.

*4) Occupancy-Centric 4-D Scene World Models:* Occupancy and 4-D scenes represent occupied, free, and evolving space. This structure supports collision checking and trajectory costs directly.

Broader 3-D generation offers representation lessons but should not be conflated with driving dynamics. GaussianCity uses a compact BEV-Point representation to generate unbounded 3-D cities efficiently [56]. Without temporal traffic evolution or action intervention, it is a scene generator rather than a driving world model under our definition.

DOME combines an occupancy VAE with spatio-temporal diffusion [40]. OccLLaMA and Occ-LLM instead tokenize occupancy with language and action [57], [58]. DynamicCity generates conditional semantic 4-D occupancy, while OccWorld forecasts occupancy and ego trajectories with open-loop planning evidence [39], [59]. Drive-OccWorld connects action-controllable occupancy forecasting to planning [60]. DriveDreamer4D, UNO, and SparseWorld explore 4-D reconstruction, continuous fields, and sparse queries [42], [61], [62].

Occupancy is planner-friendly because it exposes free and uncertain space, including occluded or non-box-shaped regions. Its drawbacks are expensive supervision and rollout. Occupancy alone also omits intent, so safety depends on how the planner uses semantics and uncertainty.

False free space is the most consequential failure for planning. Average IoU can hide small occupied regions near an ego trajectory, while over-conservative occupancy can block feasible motion. Evaluation should therefore stratify errors by distance to candidate paths and report whether probabilistic occupancy changes the chosen action or safety margin.

*5) LiDAR World Models:* LiDAR preserves sensor geometry but remains sparse. RangeLDM is static range-view synthesis; OLiDM adds object control [41], [63]. Copilot4D conditions future point clouds on ego poses (A=E), without uncertainty-aware decisions (U=P); LidarDM combines latent 4-D scenes with raycasting [38], [64]. LiDARCrafter uses text-conditioned scene graphs, and ViDAR uses point-cloud forecasting as camera supervision [65], [66].

Evaluation must cover object quality, temporal consistency, scan physics, and downstream utility. Sensor realism alone cannot establish better driving behavior because interaction and route feedback remain separate assumptions.

Beam pattern, missing returns, reflectance, boundaries, and occlusion all affect whether a generated scan is useful for perception. Range-view diffusion and latent raycasting encode different sampling assumptions. A credible study should dis-

TABLE III
REPRESENTATIVE AUTONOMOUS-DRIVING WORLD MODELS (PART 2: PLANNING WITH WORLD MODELS)

| Study | Subclass / State | World-Model Decision Interface | MPC Evidence A/C/H/U/R | Evaluation Evidence |
|---|---|---|---|---|
| MP3 [67] (CVPR'21) | Trajectory Evaluation / BEV | BEV map and agent futures feed interpretable planning costs. | P/E/E/E/E | Open-loop and closed-loop urban-driving evaluation. |
| MILE [50] (NeurIPS'22) | Latent Policy Learning / latent | Action-conditioned latent simulator supports imitation-policy rollout. | E/–/P/P/P | Closed-loop urban simulation; the world model is primarily used for learned imagination rather than online MPC. |
| DTPP [68] (ICRA'24) | MPC-style Planning / trajectory | Candidate trajectory trees condition joint prediction and learned costs. | E/E/E/–/E | nuPlan open-loop, non-reactive and reactive closed-loop planning. |
| Think2Drive [69] (ECCV'24) | Latent Policy Learning / latent | Action-conditioned latent transitions, rewards, and policy imagination. | E/E/P/P/P | Closed-loop CARLA-v2 policy evaluation; world-model rollouts primarily support training. |
| Drive-OccWorld [60] (AAAI'25) | Trajectory Evaluation / occupancy | Candidate-conditioned 4-D occupancy and occupancy-based trajectory cost. | E/E/E/P/– | nuScenes/nuScenes-Occupancy/Lyft; forecasting and open-loop planning. |
| Drive-WM [70] (CVPR'24) | Trajectory Evaluation / visual | Candidate maneuvers condition multiview futures that a separable image-reward module compares. | E/E/E/P/– | nuScenes; open-loop planning and visual-forecasting evidence. |
| AdaptiveDriver [71] (ICRA'25) | MPC-style Planning / IDM | BehaviorNet predicts reactive IDM parameters; simulated proposals are scored and executed. | E/E/E/P/E | Reactive nuPlan closed-loop evaluation with online MPC-style proposal scoring. |
| WoTE [72] (ICCV'25) | Trajectory Evaluation / BEV | Candidate trajectories query future BEV states and trajectory scores. | E/E/P/P/E | NAVSIM open-loop and reactive Bench2Drive closed-loop evaluation. |
| Raw2Drive [73] (NeurIPS'25) | Latent Policy Learning / latent | Action-conditioned aligned latent rollout for policy optimization. | E/E/P/P/P | Closed-loop CARLA-v2 policy evaluation; aligned world-model rollouts primarily support training. |
| AdaWM [74] (ICLR'25) | Latent Policy Learning / latent | Action-conditioned dynamics and reward; mismatch-guided model/policy finetuning. | E/E/P/P/P | CARLA interaction and adaptation; learned-policy execution, not online candidate MPC. |

The A/C/H/U/R rubric follows Table II; Supplementary Table S2 supplies per-entry evidence. H=E includes explicit geometric/dynamic penalties or filtering, not a hard safety guarantee. Policy-training imagination is distinguished from online rollout-based replanning. ReSim is listed in Table IV because it also implements agent–generator feedback.

close these assumptions and test the same downstream detector or planner on real and generated sequences.

### B. Planning with World Models

Planning with world models asks how imagined futures change a decision. Outputs may be rewards, values, trajectory scores, occupancy costs, or latent transitions. Unlike logged-future generation, this role must support counterfactual ego actions through candidate rollout, latent imagination, or MPC-style optimization.

Fig. 5 distinguishes Trajectory Evaluation, Latent Policy Learning, and MPC-style Planning. The subclasses depend on how the decision module consumes the model, not on state modality.

*1) Trajectory Evaluation:* Trajectory evaluators score candidate consequences. MP3 and ST-P3 combine future scene estimates with trajectory scoring [67], [75]. WoTE maps candidates to future BEV states, and Drive-OccWorld uses action-conditioned occupancy and flow [60], [72]. Drive-WM compares action-conditioned multiview futures using image rewards. ReSim also exposes a Video2Reward evaluator, but additionally implements an agent–generator feedback loop; its primary profile is therefore in Table IV [70], [76].

Table III separates costs, constraints, uncertainty, and feedback. MP3 uses map-distribution standard deviations in a speed-dependent cost (U=E). Drive-WM and Drive-OccWorld directly score geometric safety terms (H=E), but their listed evaluations do not execute replanning (R=–). DTPP filters dynamically invalid trajectories and reports reactive nuPlan planning (H=R=E); WoTE uses learned simulation rewards (H=P) and executes online evaluation in Bench2Drive (R=E) [60], [67], [68], [70], [72]. Explicit penalties, decision-usable uncertainty, and repeated execution are separate properties [77], [78].

Traceability is a practical advantage. An unsafe score can be linked to predicted occupancy, agent distance, rule violation, or a learned reward. However, no evaluator can recover a maneuver absent from the candidate set. Candidate diversity and uncertainty tests must therefore accompany gains in mean planning score.

*2) Latent Policy Learning:* Latent world models roll dynamics, reward, or value forward without reconstructing every observation. PlaNet and the Dreamer family established this approach, while MuZero, EfficientZero, and TD-MPC use learned states for search or trajectory optimization [79]–[85].

For driving, Think2Drive uses latent dynamics as a neural simulator; PIWM models individual interactions; Raw2Drive adds privileged guidance; and AdaWM adapts selectively under task or distribution shift [69], [73], [74], [86].

In Table III, Think2Drive, Raw2Drive, and AdaWM expose A/C but have partial R: imagined returns train policies, while evaluation executes the learned policy [69], [73], [74]. Their stochastic latent dynamics support U=P, without uncertainty-dependent control. AdaWM uses distribution mismatch to select model or policy finetuning; this is distinct from predictive-risk calibration. Compact states still need geometry and rule

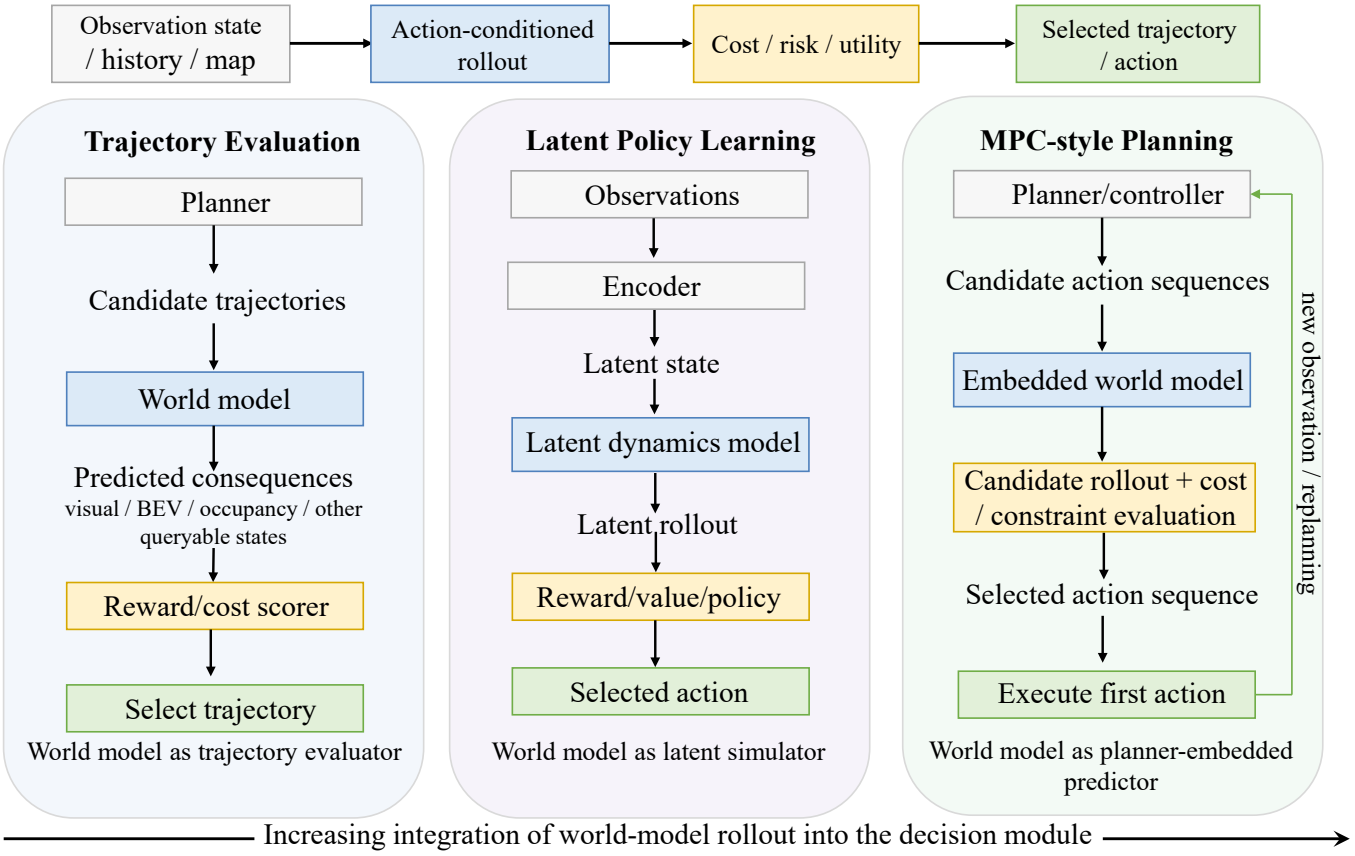



Fig. 5. Planning with world models. Rollouts enter a separable decision module through Trajectory Evaluation, Latent Policy Learning, or MPC-style Planning. These subclasses depend on how the model is used; predicted states may be visual, BEV, occupancy, vector, or latent.

anchors to preserve driving constraints [87], [88].

MILE already examines latent content through BEV semantic decoding, a no-segmentation ablation in closed-loop driving, and qualitative future BEV sequences [50]. Further quantitative tests can extend this evidence by tracking lane connectivity, occluded agents, traffic-control transitions, and collision margins across rollout horizons and matched interventions, then relating these errors to planning failures.

*3) MPC-style Planning:* MPC-style methods embed learned prediction in an explicit planner to improve rollout, scoring, or action selection.

MPC optimizes predicted behavior under dynamics, constraints, and costs over a receding horizon [13], [14], [89]. Motion-planning reviews and crash-mitigation planners similarly stress geometric feasibility, dynamic constraints, and traffic safety [77], [90]. Learning can refine the model, safe set, value, or terminal cost while retaining an optimization interface [85], [91]. The world model enriches MPC prediction rather than replacing MPC.

PDM-C shows that a structured proposal, internal-simulation, and scoring pipeline can outperform stronger open-loop fitting [92]. AdaptiveDriver learns interpretable IDM parameters from history and lane graphs [71]. DTPP is an MPC-style planning method. Its explicit tree policy planner queries jointly trained conditional prediction and cost models to select the first-stage trajectory [68].

This approach remains interpretable and close to deployed stacks, although planner design can limit candidate diversity. It also makes uncertainty operational: probabilistic occupancy, motion modes, stochastic latents, or ensembles can alter margins and fallback behavior [93], [94]. Learned interaction models can then complement classical priors that preserve constraints and interpretable costs.

The division of responsibility should be explicit. Learned transitions can model complex interaction and adapt to local traffic, while rule-based proposals and safety envelopes retain hard feasibility checks. This arrangement also supplies a fallback when the learned model is stale, uncertain, or too slow for the current replanning deadline.

### C. Hybrid Prediction–Planning World Models

Hybrid methods share prediction and action more deeply, reducing the mismatch between generated variables and planner needs. Their coupled objectives also make failures harder to attribute, so evaluation must separate representation, generation, and control effects.

Fig. 6 shows three qualifying mechanisms: a unified observation–action objective, a shared latent or structured state, and executable closed-loop interaction. The distinction from Fig. 5 is architectural. A planning-primary method retains a separable evaluator, optimizer, or policy interface; a hybrid method removes part of that separation.

TABLE IV
REPRESENTATIVE AUTONOMOUS-DRIVING WORLD MODELS (PART 3: HYBRID PREDICTION–PLANNING SYSTEMS)

| Study | Subclass / State | Joint Prediction–Decision Interface | MPC Evidence A/C/H/U/R | Evaluation Evidence |
|---|---|---|---|---|
| GenAD [95] (ECCV'24) | Shared structured latent | Instance-centric future and ego-plan generation share scene tokens. | P/P/P/P/– | nuScenes; open-loop end-to-end planning metrics. |
| DrivingGPT [96] (ICCV'25) | Unified AR tokens | Observations and actions are modeled in one autoregressive sequence. | E/–/P/P/– | nuPlan/NAVSIM planning and world-modeling evidence. |
| World4Drive [97] (ICCV'25) | Shared physical latent | Intention-conditioned latent futures support multimodal trajectory selection. | E/E/P/P/– | nuScenes/NAVSIM; open-loop planning and non-reactive simulation metrics. |
| DriveArena [98] (ICCV'25) | Interactive simulator | Ego actions update a generative traffic world with simulator feedback. | E/P/P/P/E | Reactive closed-loop simulation and driving evaluation. |
| DriveLaW [99] (arXiv'25) | Shared visual latent | Video latents condition a diffusion action planner; motion is encoded as text. | P/–/P/P/– | Planning benchmarks and video-generation evaluation. |
| UniDrive-WM [100] (arXiv'26) | Unified VLA tokens | Language, vision, and action tokens support understanding, planning, and generation. | E/P/P/P/P | nuScenes and reactive Bench2Drive policy execution; no online candidate-rollout optimization. |
| WorldDrive [101] (arXiv'26) | Shared vision–motion state | Trajectory-conditioned generation, inherited planning representations, and future-aware reward. | E/E/P/P/– | nuScenes, NAVSIM, and NAVSIM-v2; generation and non-reactive planning evidence. |
| DriveWorld-VLA [102] (arXiv'26) | Shared VLA/BEV latent | Action-conditioned BEV imagination and learned reward refine action training. | E/E/P/P/– | nuScenes/NAVSIM-v1/v2; open-loop and non-reactive planning, not reactive execution. |
| VectorWorld [55] (arXiv'26) | Interactive vector/graph simulator | Lane–agent generation and reactive traffic; ego execution updates the scene and feedback. | E/P/E/P/E | nuPlan/Waymo generation; Waymo closed-loop planner evaluation and physics violations. |
| ReSim [76] (NeurIPS'25) | Interactive visual simulator | Action-conditioned video and Video2Reward; generated frames feed back to a driving agent. | E/E/P/P/E | Non-reactive NAVSIM selection; model-only visual feedback demonstrated from two seeded scenes. |

The rubric follows Table II; Supplementary Table S3 gives per-entry evidence. Hybrid denotes shared state/objective or an executable interaction loop. ReSim has model-only visual feedback; DriveArena steps a traffic simulator. VectorWorld H=E concerns non-ego kinematic shaping, not guaranteed ego safety. Stochastic futures or mode scores alone receive at most U=P.

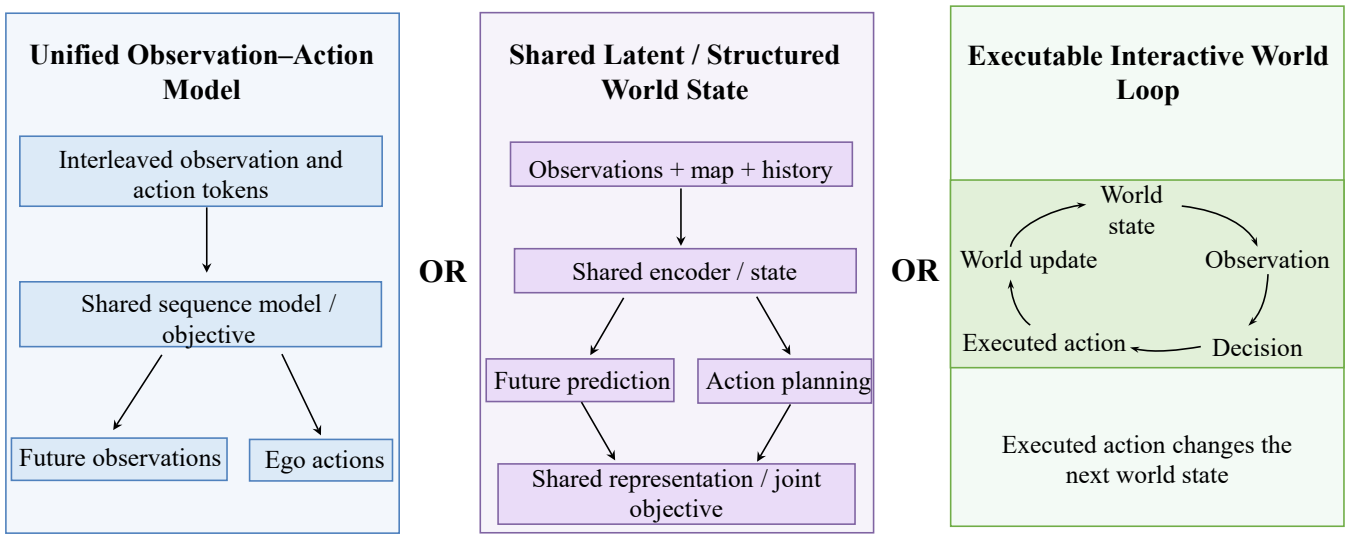



Fig. 6. Qualifying mechanisms for hybrid prediction–planning world models. Hybrid systems use at least one joint prediction–decision objective or unified observation–action interface, shared latent or structured state, or executable closed-loop interaction. A separable generated-future evaluator alone remains in the planning category.

*1) Unified Observation–Action World Models:* DrivingGPT interleaves image and action tokens under a next-token objective [96]. This provides action-conditioned generation, but the generic MDP reward definition does not establish an implemented cost/reward interface (C=–). Drive-WM instead feeds generated videos to a separable image-reward module and remains planning-primary. ReSim qualifies as an interactive hybrid through its implemented agent–generator feedback loop

[70], [76].

Unified tokens can capture visual consequences omitted by hand-crafted costs. They also couple failure modes: video artifacts can alter actions, and plausible frames do not ensure feasible control.

*2) Unified Latent Driving World Models:* GenAD uses an instance-centric latent for agent evolution and ego trajectories, while World4Drive selects intention-conditioned latent futures [95], [97]. WorldDrive unifies trajectory-conditioned scene generation and planning through shared vision–motion representations and a future-aware rewarder [101]. It differs from the earlier Generalized Predictive Model, which studies predictive pretraining and transfer [103]. DriveLaW, UniDrive-WM, and DriveWorld-VLA further connect generative latents, reasoning, and actions [99], [100], [102].

Table IV distinguishes shared states from cost and feedback interfaces. GenAD has indirect training-time cost/constraint support (C=H=P); World4Drive exposes ScoreNet (C=E), while intention scores alone give U=P [95], [97]. DriveWorld-VLA adds action-conditioned BEV imagination and reward-guided refinement (A=C=E), without reactive execution evidence (R=–) [102]. UniDrive-WM reports reactive Bench2Drive policy execution (R=P), distinct from online candidate-world-model optimization [100]. Shared representations do not by themselves implement feasibility or uncertainty-aware control.

Evaluation should isolate the source of each gain. Generation fidelity, latent consistency, action quality, and closed-loop stability need separate ablations even when trained jointly. Without them, a higher driving score cannot show whether the shared representation improved prediction or whether the planner simply learned around its errors.

*3) Closed-loop Interactive World Simulators:* Interactive simulators couple agents to a world update. DriveArena combines a traffic manager and generated observations [98]. ReSim feeds back model-generated video; VectorWorld steps structured states and shapes non-ego kinematics (H=E). Both have R=E within these distinct loops [55], [76]. DrivingSphere uses 4-D occupancy composition/rendering; RealEngine and UniSim integrate reconstruction, traffic, and neural rendering [104]–[106].

Their value is executable feedback after the ego vehicle changes the scene. Credibility, however, depends on state construction, rendering, background-agent behavior, and protocol design. Waymax, ScenarioNet, Nocturne, and SimNet show how logged scenes can seed multi-agent rollout [107]–[110]. World-model simulators should report intervention robustness, agent reactivity, coverage, and reproducible failures, not only visual quality or mean driving score.

Reactive evidence requires more than looping generated frames. Background agents must respond to the ego action, maps and traffic lights must update consistently, and collisions must have repeatable semantics. Fixed seeds and scenario-level failure logs are needed so that a planning regression can be reproduced rather than absorbed into one aggregate score.

## IV. Cross-Paradigm Comparison

Motivated by the receding-horizon logic of MPC, this section compares the information that each world model provides to a planner that repeatedly predicts, scores, constrains, and updates ego actions.

### A. Mechanism–Task Matching

Mechanisms should be compared by task role, not ranked as generators. Diffusion suits high-dimensional futures when controllability and perceptual quality matter [40], [43], [111]. Autoregressive transformers provide token interfaces for video, text, action, occupancy, and object-centric BEV states [35], [49], [57], [112]. Latent dynamics fit rollout and value estimation because repeated imagination is cheap [69], [80], [85]. Hybrid designs connect prediction and planning, but complicate optimization and evaluation.

Control profiles differ. Diffusion captures multimodality, although sampling is costly and constraints can drift. Autoregressive models tokenize many modalities but accumulate sequential errors. Latent dynamics support cheap imagination; their compact states are harder to verify. Graph models preserve relations and maps while omitting some appearance and dense geometry. The practical choice is the simplest representation that retains variables used by the downstream decision.

### B. Cross-Representation Trade-offs

Visual states support rendering and inspection, but planning requires an additional interface. BEV offers compact geometry at the cost of vertical detail. Occupancy exposes free space directly, although dense supervision and rollout are expensive. LiDAR retains metric sensor structure but remains sparse and difficult to generate. Latent states can support compact rollout, while their semantic content requires separate validation. In MPC terms, representations differ in how directly they expose optimizer states, constraints, and costs.

Supervision changes the trade-off. Images are abundant but weakly structured. BEV and occupancy labels require detection, alignment, fusion, or annotation, while LiDAR supplies geometry without intent. Latent states may fail silently when their objectives omit safety-critical variables. Uncertainty must be actionable: samples, cell probabilities, trajectory modes, or stochastic latents should alter speed, margins, yielding, or fallback behavior. The MPC lens asks whether uncertainty changes the feasible set, safety margin, terminal condition, or candidate cost.

### C. Rollout Cost Across Paradigms

Rollout work depends on the representation consumed by the decision module. Drive-WM generates multiview futures for image-based reward evaluation [70]. Think2Drive learns its actor–critic from imagined latent states and predicted rewards [69]. Image-based scoring includes generating and evaluating observations, whereas direct latent reward prediction can bypass image decoding. These paths expose different generation, decoding, and scoring workloads.

TABLE V
PUBLISHED QUANTITATIVE EVIDENCE GROUPED BY EVALUATION SETTING. HIGHER IS BETTER FOR PDMS, DS, AND SR; LOWER IS BETTER FOR $L_2$ AND COLLISION RATE (CR). NUSCENES IS OPEN LOOP; NAVSIM USES NON-REACTIVE LOGGED TRAFFIC; BENCH2DRIVE IS REACTIVE SIMULATION.

| Protocol group | Method | Sensor input | World-model role | Reported result | Permitted interpretation / constraint |
|---|---|---|---|---|---|
| nuScenes val, TemAvg | GenAD [95] | C | Joint ego/agent trajectory latent | $L_2$ 0.52 m; CR 0.19% | Dagger row; 2-s ego history and map/box/motion supervision. |
| nuScenes val, TemAvg | Drive-OccWorld$^P$ [60] | C | Occupancy rollout and scoring | $L_2$ 0.47 m; CR 0.11% | TemAvg without extra ego status; predicted trajectory conditions. |
| NAVSIM test set | World4Drive [97] | C | Intention-conditioned physical latent | PDMS 85.1 | Camera only; same split/PDMS as next row. |
| NAVSIM test set | WoTE [72] | C+L | BEV rollout and scoring | PDMS 88.3 | Sensors, training, and candidates differ; system-level comparison. |
| NAVSIM navmini | DrivingGPT [96] | front-C | Joint AR visual/action tokens | PDMS 82.4 | Separate navmini split, distinct from full test-set PDMS. |
| Bench2Drive, 220 routes | WoTE [72] | C | BEV scoring in a TCP agent | DS 61.71; SR 31.36% | Reactive CARLA; same route set as next row. |
| Bench2Drive, 220 routes | Raw2Drive [73] | multi-C+IMU | Dual-stream latent RL | DS 71.36; SR 50.24% | Training/policy differ; system-level, not isolated model effects. |

C: camera; front-C: front-view camera; L: LiDAR; IMU: inertial measurement unit; PDMS: Predictive Driver Model Score; DS: Driving Score; SR: Success Rate. nuScenes values average 1/2/3-s columns after temporal aggregation. Bench2Drive uses 220 short routes.

GAIA-1 combines autoregressive token prediction with a diffusion video decoder [35]. Token count and denoising steps therefore contribute separate work budgets, alongside candidate count $N$, horizon $H$, and stochastic samples. Comparisons should fix resolution, hardware, precision, batching, and rollout budgets, and time encoding, rollout, decoding, and scoring separately. For Think2Drive, training-time imagination must also be distinguished from deployed policy inference. Section V-F specifies the corresponding runtime evaluation requirements.

### D. Utility-Oriented Comparison

Generation quality and driving utility are not equivalent. FID and FVD measure distributional realism, not geometry, control following, or planning improvement [113], [114]. Utility appears when the model changes rollout, costs, constraints, or replanning. WoTE, AdaptiveDriver, and AdaWM are less visually direct but closer to trajectory selection and adaptation [71], [72], [74]. Bench2Drive exposes closed-loop failures that open-loop metrics miss, while NAVSIM provides a non-reactive middle ground [17], [18].

Table V adds nuScenes validation results from the published GenAD and Drive-OccWorld$^P$ TemAvg rows [60], [95]. TemAvg averages 0.5 s to each 1/2/3-s horizon; values are the source Avg columns. This group excludes NoAvg endpoint metrics, extra-ego-status Drive-OccWorld variants, and ground-truth-conditioned upper bounds. Inputs, supervision, and planners differ, limiting interpretation to reported systems. Results across dataset/split groups are non-comparable.

Read across Tables II–IV, Copilot4D has an action-conditioned forecast (A=E) but no uncertainty-dependent decision rule (U=P). DrivingGPT likewise exposes A without an implemented C interface. MP3 provides a concrete contrast: map uncertainty changes a planning cost (U=E). Policy-learning methods retain partial R despite explicit A/C, while ReSim, DriveArena, and VectorWorld expose feedback at different simulator levels. These profiles compare functions and evaluation scope; counting E entries would conflate controllability, constraints, uncertainty use, and execution fidelity.

## V. DATASETS, SIMULATORS, EVALUATION, AND BENCHMARKS

### A. Training Datasets

Here, datasets are training resources rather than evaluation protocols. DrivingDojo supports video synthesis and conditioned generation [115]. KITTI, nuScenes, Waymo, and nuPlan provide calibrated sensors, maps, poses, and trajectories for structured modeling [16], [116]–[118]. Occ3D, OpenOccupancy, Cam4DOcc, and UniOcc add dense 3-D or 4-D supervision [119]–[122]. The key issue is coverage matching: data must supply the modality, annotations, and temporal context required by the target state.

Requirements differ by state. Video needs variation in appearance and ego motion; BEV and occupancy need calibration, pose, maps, and geometric labels. Planning needs routes and expert behavior, whereas LiDAR needs aligned scans and sensor-consistent geometry. Argoverse, Waymo Open Motion, and Lyft Level 5 remain useful for interaction and map-conditioned futures [123]–[126]. Dataset reports should describe scenario distribution, route diversity, interaction difficulty, and annotation quality, not size alone.

### B. Simulation Platforms and Ecosystems

Simulation platforms determine how models interact with an environment. CARLA provides configurable sensors and closed-loop execution [127]. MetaDrive and LimSim support scalable scenarios and traffic simulation [128], [129]. Waymax initializes multi-agent scenes from logs and adds route-aware states, reactive agents, and closed-loop metrics [107]. Learned ecosystems also generate observations, turning a predictor into a simulator component.

Classical simulators offer repeatable physics and explicit rules but limited visual and behavioral diversity. Learned simulators inherit real-log statistics, yet interventions far from the data manifold remain unreliable. Hybrid systems combine explicit traffic structure with learned rendering or behavior. Each platform should disclose its state, route, intervention interface, metrics, and background-agent assumptions.

### C. Evaluation Metrics

FID, FVD, PSNR, SSIM, and human preference assess visual similarity [113], [114]. Occupancy/point clouds need geometric, temporal, and object-level tests; conditioned generation needs control-following tests. nuScenes $L_2$ measures displacement from logged ego trajectories; CR measures collisions with logged traffic, giving partial level (4) evidence in open loop. Driving needs progress, comfort, compliance, and closed-loop safety. Displacement error can correlate weakly with driving performance [130]; risk-aware planning needs safety margins under interaction [77], [78], [94].

For safety-critical generation, fidelity should be decomposed into five testable levels. (1) *Conditioning fidelity* tests whether the generated ego motion follows an emergency brake or avoidance command. (2) *Kinematic and geometric fidelity* tests acceleration, jerk, yaw rate, free space, and collision geometry. (3) *Interaction fidelity* tests whether surrounding agents react causally to the intervention. (4) *Risk fidelity* uses time-to-collision, post-encroachment time, minimum distance, and collision severity rather than appearance alone. (5) *Coverage and calibration* tests long-tail category frequency and severity without equating synthetic frequency with real-world risk. A model that improves average FVD or displacement error may still miss rare cut-ins, occluded pedestrians, emergency vehicles, or construction zones. Tail-sensitive metrics, scenario strata, uncertainty intervals, and failure analysis are therefore necessary complements.

The metrics in Table V cover parts of this framework. NAVSIM's Predictive Driver Model Score (PDMS) multiplies no-at-fault-collision and drivable-area-compliance factors by a weighted average of ego progress, time-to-collision, and comfort [18]. Comfort and road/collision geometry provide level (2) proxies, while time-to-collision and collision penalties contribute level (4) evidence. Bench2Drive's Driving Score (DS) averages route completion multiplied by infraction penalties. Its Success Rate (SR) is the fraction of routes completed within the time limit without traffic violations [17]. DS and SR summarize task success and compliance, with collision-related outcomes providing partial level (4) evidence.

These are planner-outcome measures, so their scores do not directly validate generated-world fidelity. PDMS uses non-reactive traffic and cannot test causal agent responses at level (3). Reactive Bench2Drive permits interaction testing, but aggregate DS/SR do not isolate response fidelity. None of these aggregate scores establishes commanded-maneuver adherence at level (1), long-tail coverage or risk calibration at level (5), or complete coverage of levels (2) and (4). Component scores and scenario strata should therefore accompany matched-command tests, intervention-based agent-response checks, and coverage/calibration analysis.

### D. Benchmark Protocols

Benchmark protocols define the evaluation contract: task, split, allowed inputs, simulator assumptions, metrics, and reporting rules. nuPlan combines planning data with reactive simulation and planning metrics [16]. Bench2Drive standardizes routes, interactions, and driving skills [17]. NAVSIM is a scalable non-reactive middle ground in which background actors follow logged futures, so it does not establish reciprocal traffic response [18]. ACT-Bench directly tests whether a generated future follows a specified trajectory condition. WorldSimBench instead separates perceptual assessment from action-level usefulness [19], [46]. Shared protocols matter because private settings make heterogeneous outputs difficult to compare.

The protocol determines what progress is visible. Log replay isolates prediction under fixed futures. Non-reactive simulation measures repeated ego execution with limited interaction evidence. Reactive simulation tests reciprocal behavior but depends on background-agent realism. Scenario-generation benchmarks expose controllability and rare events, not necessarily everyday driving.

Cross-paper comparison also needs a protocol identity. Results belong to the same comparison group only when the benchmark split, permitted sensors, route set, agent-reactivity assumption, and metric implementation match. Training data and planner design may still differ, so such values remain system-level evidence. When any protocol field changes, results should be reported in separate groups rather than merged into a single ranking.

For emergency avoidance and long-tail events, a credible protocol should pair rollouts from the same initial state under distinct feasible interventions. It should verify execution of each conditioning maneuver, stratify results by conflict type and severity, and report fixed seeds for stochastic generation. Each rollout should pass the five fidelity levels above before its downstream planner outcome is counted. This counterfactual design distinguishes an intervention-sensitive world model from one that merely produces plausible-looking hazards. Future benchmarks should state their evidence target clearly: representation quality, actionability, planning improvement, interaction robustness, safety validation, or scenario-generation capability.

### E. Open-Loop and Closed-Loop Evidence

Open-loop evaluation is cheap, scalable, and reproducible, but measures agreement with logged futures under fixed distributions. Once a world model enters a planner, its predictions affect ego actions and future observations. The resulting open-loop–closed-loop gap has five mechanisms. *Observational domain shift* occurs when deployed states depart from logged inputs. *Objective mismatch* occurs when imitation or reconstruction losses omit closed-loop consequences. *Interaction mismatch* occurs when background agents remain non-reactive. *Compounding error* accumulates state and execution errors over a horizon. *Interface mismatch* occurs when model uncertainty or constraints cannot alter the planner.

These mechanisms require distinct remedies rather than one aggregate "closed-loop score." Adaptation or data aggregation can reduce observational shift. Decision-aware and multi-step objectives can reduce objective and temporal mismatch. Reactive-agent simulation can expose interaction failures, and receding-horizon updates can limit accumulation. Calibrated uncertainty, cost, and constraint interfaces can make model limitations actionable [20].

Evidence should consequently form a ladder. Open-loop tests isolate prediction, geometry, calibration, and counterfactual controllability. Non-reactive simulation tests repeated ego execution at scale but cannot establish reciprocal interaction. Reactive closed-loop simulation tests policy–environment feedback. On-road or high-fidelity replay validation checks whether conclusions survive simulator assumptions. A claim should not be transferred upward without the corresponding experiment. Balanced evaluation combines these levels and reports where improvement disappears, treating open- and closed-loop results as complementary evidence rather than a valid/invalid dichotomy.

### F. Runtime and Safety Evaluation Requirements

Runtime evaluation should specify the sensing timestamp, planning deadline, and action taken after a deadline miss or detected shift. Report median and tail latency, memory, replanning rate, candidate/sample budgets, horizon, hardware, precision, and batching alongside driving utility. Test fallback behavior, regressions after model updates, and whether monitors detect failures before action. These are additional tests for online use; benchmark scores remain distinct from system-level safety assessment under ISO 26262, ISO 21448, or UL 4600 [131]–[133].

## VI. Open Challenges and Future Trends

The five challenges in Fig. 7 form a dependency chain. Representation determines which geometry, agents, semantics, and uncertainty survive long-horizon rollout. Rollout quality and planner-facing interfaces determine whether generation changes a decision. Computation limits the number of futures queried before a deadline. Foundation models broaden semantic coverage but add grounding and latency risks. Deployment and safety validation depend on all four upstream properties. Progress should therefore report state validity, horizon, decision utility, latency, and safety evidence together.

### A. Long-Horizon Reliability

Long-horizon reliability asks whether imagined futures remain stable enough for safe decisions. More generated frames are not sufficient. Errors accumulate as temporal drift, trajectory deviation, and reduced safety margins; non-reactive agents can hide interaction failures. Protocols should vary rollout length, replanning rate, feedback, agent reactivity, calibration, and constraints. They should report when each metric becomes unreliable. One test is a horizon-conditioned curve linking rollout error to safety-margin error. Another compares one shared transition with specialized short, interaction, and route-level models under the same closed-loop protocol.

Failure thresholds should be tied to the consumer. A small pixel error may be harmless, while a one-cell free-space error near the planned path can reverse a decision. Horizon studies should therefore report both state error and the first rollout step at which the planner's selected action diverges from a trusted reference.

### B. Unified Multimodal World Representation

Driving states combine appearance, 3-D geometry, maps, interactions, actions, and sometimes language. Shared BEV, occupancy, or multimodal latents seek one abstraction for prediction and planning. Feature alignment alone does not ensure geometric, causal, or control consistency. A direct test perturbs appearance, sensors, and language while holding topology and collision geometry fixed, then checks whether costs and constraints remain invariant. Selective-sharing experiments should also compare fully shared and task-specific subspaces with reconstruction and closed-loop metrics. Language and VLA models add semantics but introduce hallucination, latency, grounding, and verification risks [12], [134], [135].

Embodied multimedia extends static sensor fusion toward a perception–decision–action loop across acquisition, communication, cognition, and evaluation [136]. For driving, such signals matter only when they preserve physical semantics and improve action. The invariances should be explicit. Appearance can vary while topology and collision geometry remain fixed; intent can vary within kinematic limits. Language describes scenarios, but physical feasibility must filter them. DriveLM, DriveVLM, and DiLu illustrate language-grounded or knowledge-driven reasoning [137]–[139].

### C. Bridging Generation, Planning, and Evaluation

Partial H for the shared-state methods in Table IV, together with partial U for Drive-OccWorld and WoTE in Table III, motivates explicit ego feasibility enforcement and decision-usable uncertainty. VectorWorld makes non-ego kinematic handling explicit; its scope differs from ego safety. A controlled study can hold the transition (A), candidate set, and cost (C) fixed, then vary constraint handling (H), uncertainty-dependent margins (U), and replanning interval (R). Violations, recovery after disturbances, and deadline misses under paired interventions would reveal which changes improve execution. The evidence ladder in Section V specifies the validation level for each conclusion.

### D. Efficiency, Scalability, and Deployment

World models must support online inference and repeated candidate rollout, sometimes inside a closed-loop simulator. Dense states, diffusion, long token sequences, and language modules increase latency and memory. Compact states reduce cost but can discard safety-relevant structure. Compression should therefore be judged by deadline compliance and any loss in geometry, calibration, constraints, or safety. Candidate techniques include structured latents, distillation, quantization, token reduction, sparse activation, and event-triggered computation. The useful target is an end-to-end latency–utility

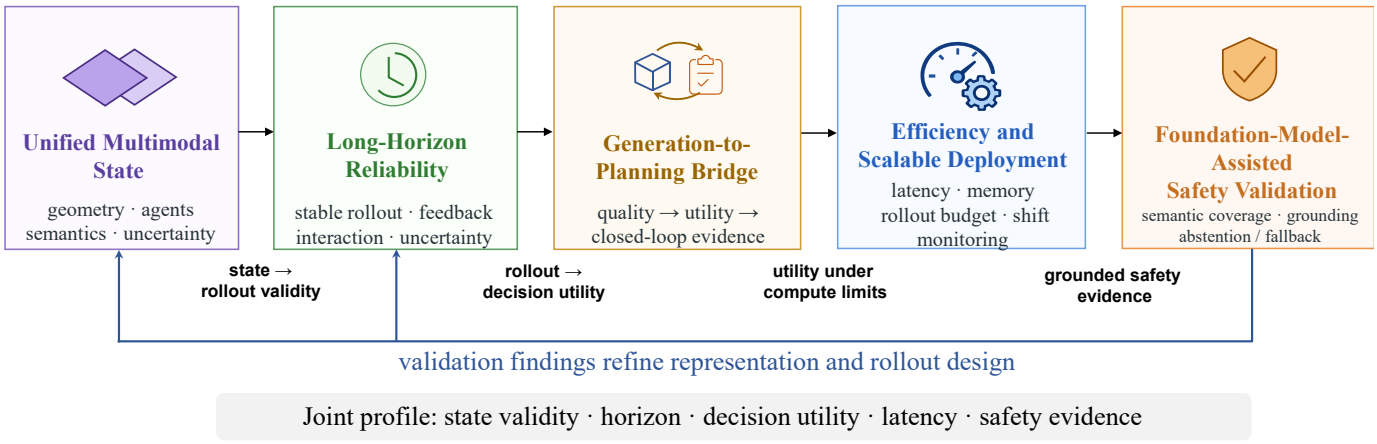



Fig. 7. MPC-oriented future challenges for autonomous-driving world models. The five directions form a dependency chain from unified multimodal state and reliable long-horizon rollout to decision utility, scalable deployment, and grounded safety evidence. Validation findings feed back to representation and rollout design, and progress should be reported as a joint profile of state validity, horizon, decision utility, latency, and safety evidence.

frontier at fixed hardware, horizon, candidate count, and safety metrics.

Fleet and domain scaling adds another constraint. Adapters can be efficient, but uncontrolled online updates weaken reproducibility and safety assurance. Future systems should separate a frozen, regression-tested safety path from updateable predictive components. They should monitor shift, uncertainty, deadline misses, and fallback activation, while preserving versioned evidence for every configuration. This design cannot replace model validation, but it can turn silent failure into detectable system behavior.

### *E. Foundation Models and Multimodal Decision Making*

Foundation models add language-grounded reasoning, scenario construction, and multimodal decisions. Their roles remain distinct. Encoders identify open-vocabulary agents or rules; reasoning interfaces form candidate decisions; generators specify rare conditions; VLA policies emit actions. None alone supplies an action-conditioned transition. Semantic alignment becomes decision evidence only when grounded in geometry and when it changes a trajectory, cost, constraint, uncertainty estimate, or fallback rule [140].

DriveLM and DriveVLM illustrate semantic reasoning and hierarchical planning. DriveWorld-VLA shares world-model latents with a VLA planner, relating actions to imagined scene evolution [102], [137], [138]. Language supplies intent and open-vocabulary priors; structured states provide physical variables. The transition supplies consequences, and the controller supplies actions. Joint training does not make their evidence interchangeable. A language explanation is not proof of physical prediction or safe control.

Failure modes include hallucinated entities or rules, weak spatial grounding, prompt sensitivity, stale context, and excessive latency. Validation should pair language and geometry perturbations, test explanation–action consistency, measure calibration and deadlines, and check whether a safety monitor rejects infeasible outputs. Generative testing can expand rare-scenario coverage. Before supporting a safety claim, however, synthesized cases must pass the five fidelity levels in Section V [141]–[144].

A concrete protocol should cross language perturbations with matched geometry. It can paraphrase a valid instruction, insert a misleading prompt into the same scene, and alter geometry while holding language fixed. Action consistency, violations, confidence, and monitor interventions then separate linguistic robustness from physical grounding. A red-team benchmark should also require abstention or fallback under contradictory instructions, unseen traffic-control semantics, or missing sensor evidence. Fluent explanations alone should not receive credit.

## VII. Conclusion

This survey studies autonomous-driving world models through an MPC lens. Future world generation rolls the state forward. Planning-oriented models convert imagined consequences into decision variables. Hybrid systems share representation, objective, or execution between prediction and action. Each role makes different trade-offs among realism, geometry, controllability, utility, and efficiency.

The comparison separates action-conditioned generation (A) from decision costs (C), while constraint handling (H), decision-usable uncertainty (U), and replanning (R) require evidence at the relevant agent and execution level. Shared prediction–action states leave ego feasibility partly addressed, even when a simulator explicitly constrains non-ego motion. Future work should test feasibility checks and uncertainty-aware repeated execution under matched protocols.

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